

Joseph Brijs

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EDUCATION

Utah State University

Masters of Science: Computer Science

Expected May 2026

Utah State University

Bachelor of Science: Computer Science

May 2025

GPA: 3.7

EXPERIENCE

Software Developer and Research Assistant

Utah State University ASPIRE Lab

Nov 2024 - Present

Logan, Utah

- Developed a Rust program to optimize charger efficiency at a Utah Transit Authority electric bus depot, reducing peak energy demand and expected to lower energy costs by 25%.
- Engineered an EV charger simulator in Rust, enhancing testing workflows and simulation fidelity for both the lab and external partners.
- Performed data analysis and trained ML models for predicting in-route electric bus energy consumption.
- Designed and implemented features for an EV Bus Depot Reliability Web app in ExpressJS and Vue, increasing charger and bus visibility and reliability for UTA depot operators

Software Engineer - Intern

Elektrik

May 2024 – Nov 2024

Lincoln, Utah

- Built a Rust/Angular search interface for electrical accessories which contributed to a 25 % increase in revenue.
- Reduced credit application time from days to minutes by creating an adaptive form, automating workflows, generating PDFs, and integrating the process into the web app dashboard
- Aided in the transition from Angular to React, reducing the frontend codebase by 40%, improving UI/UX, and enhancing developer efficiency which resulted in a more maintainable codebase

PROJECTS

Inbox AI | *Go, PostgreSQL, Ollama, React.js, Docker Compose*

- Built an AI email pipeline with Go API, Postgres (pgvector), and Ollama LLMs for classification and summarization
- Designed Postgres schema with automatic migrations and indexing for entity-relationship mapping
- Implemented email ingestion with Gmail OAuth, background jobs, and rate limiting for reliable sync
- Developed a RAG system using nomic-embed-text embeddings for sub-second semantic search
- Orchestrated infrastructure with Docker Compose (migrations, health checks, service dependencies)
- Created a React dashboard with semantic search, filtering, and knowledge graph visualization

Stock Price Neural Network Web App | *Python, PyTorch, Flask, React.js, AWS, Docker*

- Built an automated data pipeline in Python to ingest daily stock prices and engineer features, stored in Amazon S3
- Trained LSTM networks (PyTorch) to predict prices for major tech equities, achieving an average RMSE of 2.5%.
- Orchestrated serverless inference: scheduled AWS Lambda jobs to fetch new data and run predictions.
- Exposed real-time predictions and historical data through RESTful API endpoints (Lambda + API Gateway).
- Developed a React.js frontend for data visualization, hosted on S3/CloudFront with DNS via Route 53 (faangfinance.com).

Spotify Genre Clustering (K-Means) | *C++, OpenMP, MPI, Python*

- Implemented a serial C++ version of K-Means to cluster 1.2 million Spotify tracks by genre.
- Parallelized K-Means with OpenMP and MPI in C++, ensuring identical results to the serial implementation.
- Developed a Python visualization script to project high-dimensional clusters into a 3D feature space.
- Conducted strong and weak scaling studies on University of Utah CHPC systems

SKILLS

Technical Skills: Rust, JavaScript, Typescript, Go, Python, Java, C++, HTML/CSS, React, Angular, Vue, Flask, AWS, Docker, Git, PostgreSQL, SQL, MongoDB, SQLite, PyTorch, pandas, Redux, Express.js, Node.js, OpenMP, MPI, Linux